

SWIFT UAV

Structural Systems v0.2.3



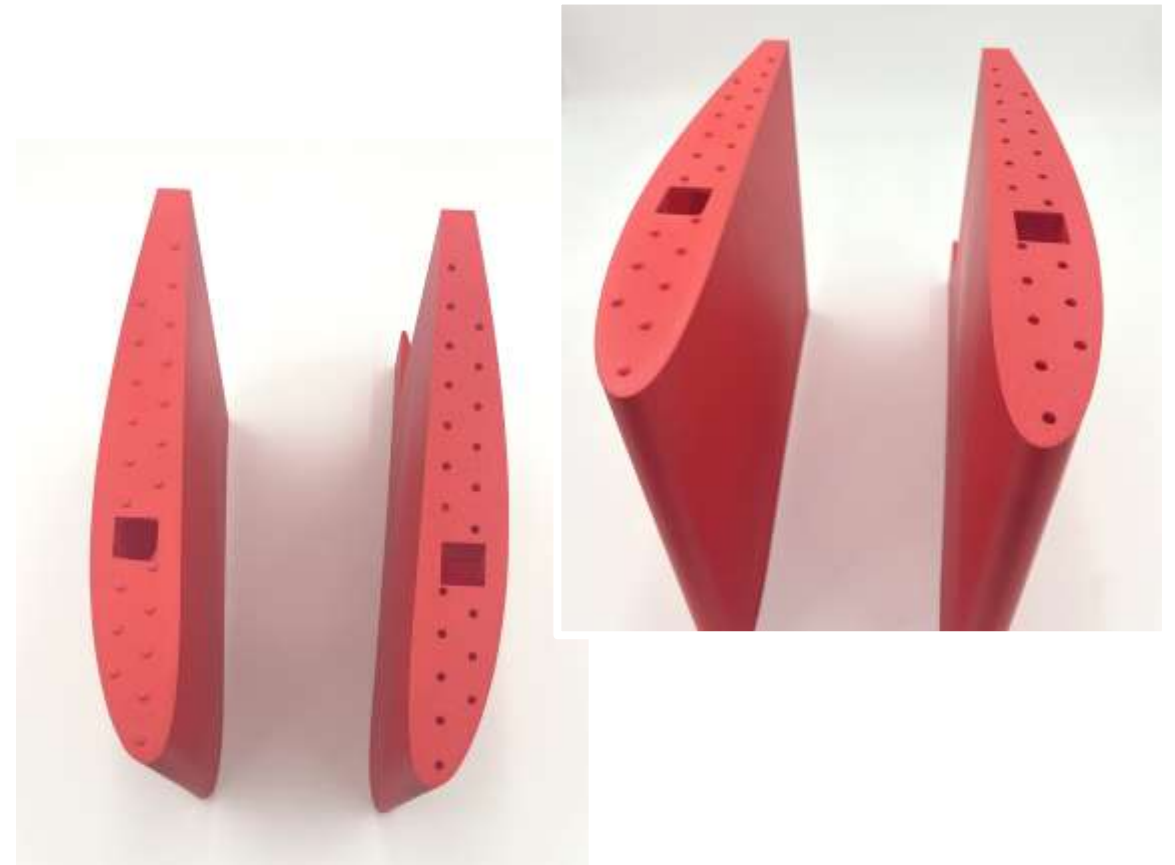
Design & Manufacturing Progress

- Current design iteration is focused on:
 - **Structural improvements**
 - **Operational improvements**
 - **Design for manufacturability**
- The current (v0.2.3) UAV design is **in production**
- Initial prints confirm the validity of design changes



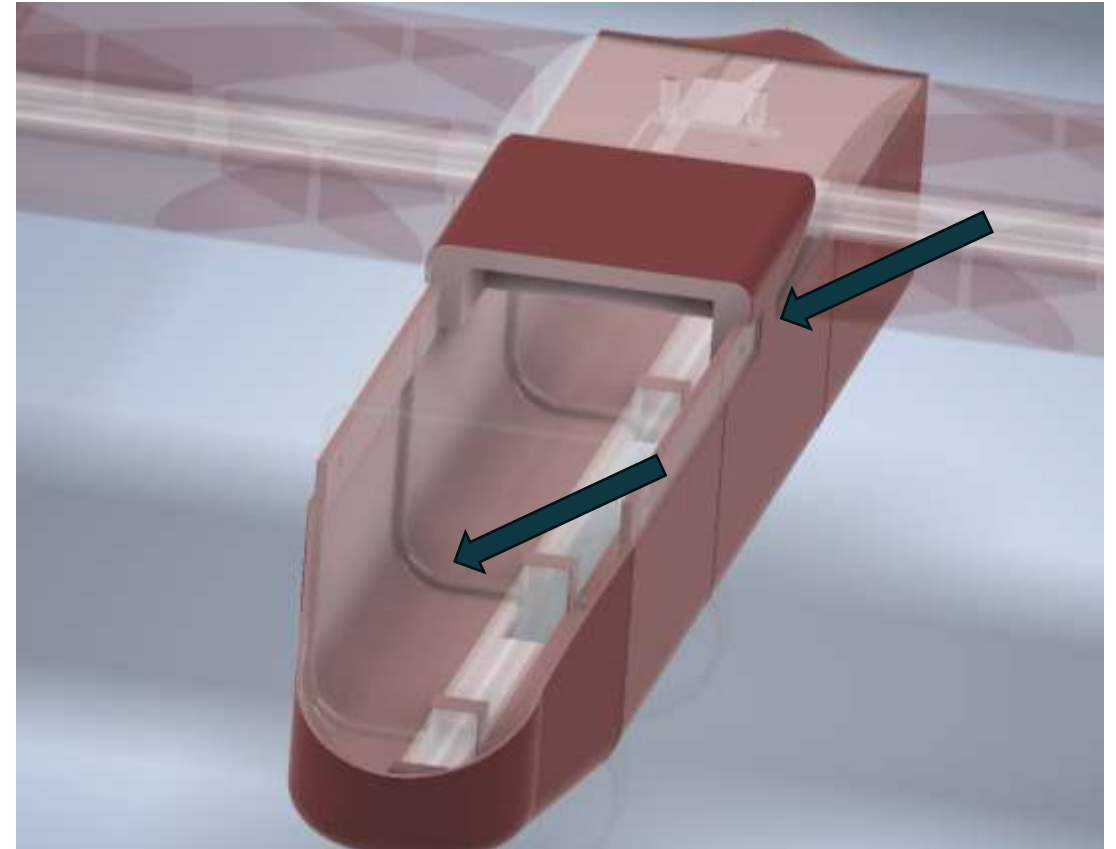
Structural Modifications for Assembly

- Locking nubs have been added between wing segments
- They provide multiple structural and functional benefits:
 - **Load Transfer:** Enables transfer of shear loads between wing sections
 - **Improved Assembly:** Enables part indexing for easy assembly
 - **Torsional Stiffness:** Increases torsional resistance substantially



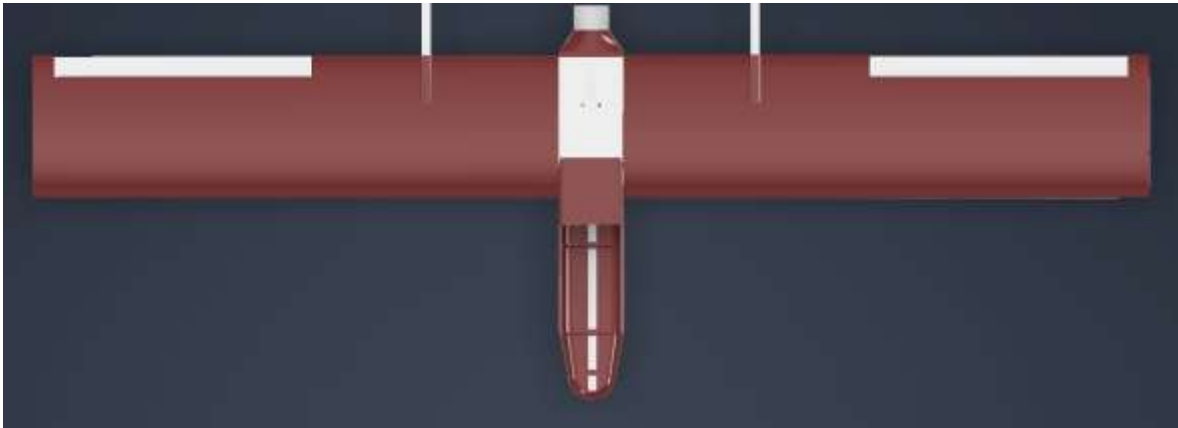
Structural Modifications for Assembly

- Tabs have been added to fuselage sections
 - Provides indexing points to **streamline manufacturing** and increase joint rigidity
- A slot has been added to the fuselage cover
 - Makes for **easier and more ridged assembly**
- A 2020 rail has been added in the fuselage to:
 - Reinforce the fuselage
 - Act as a mounting point for the landing gear
 - **Act as a hardpoint for payloads**



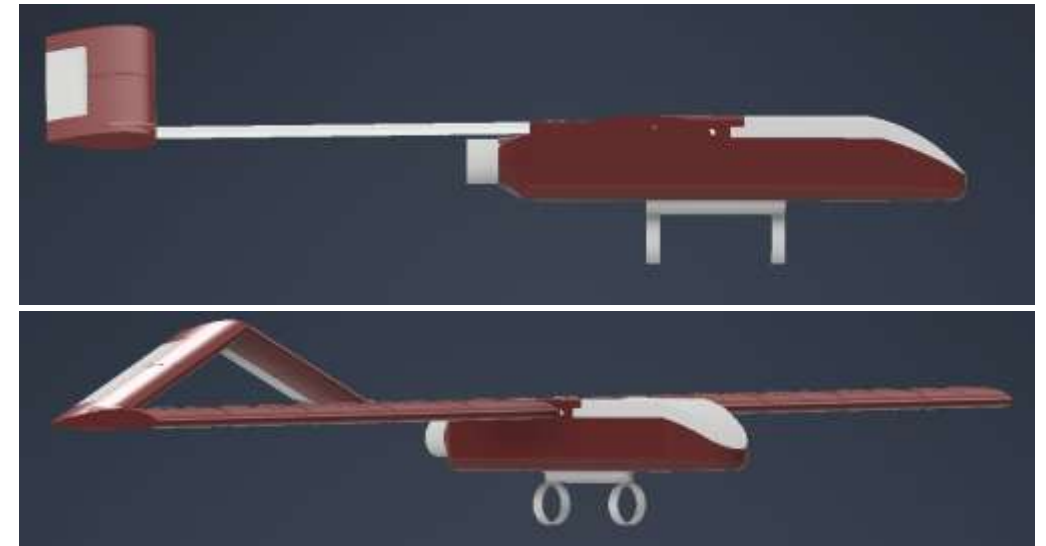
Avionics Integration

- Preliminary Avionics integration is being conducted in parallel with the manufacturing
 - Ensuring that mounting points and clearances for all the hardware correct
- The 2020 aluminum bar embedded into the fuselage base also serves:
 - A **universal mounting interface** for avionics trays and payloads.
 - And enables **easy reconfiguration** of internal loadout for missions

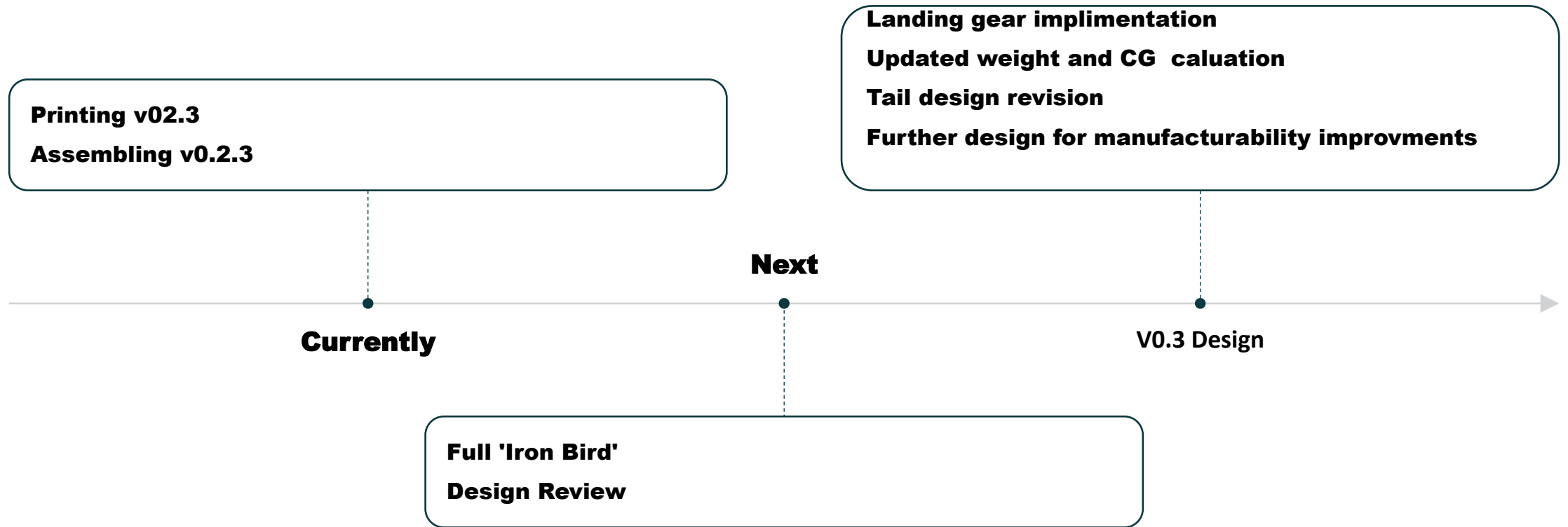


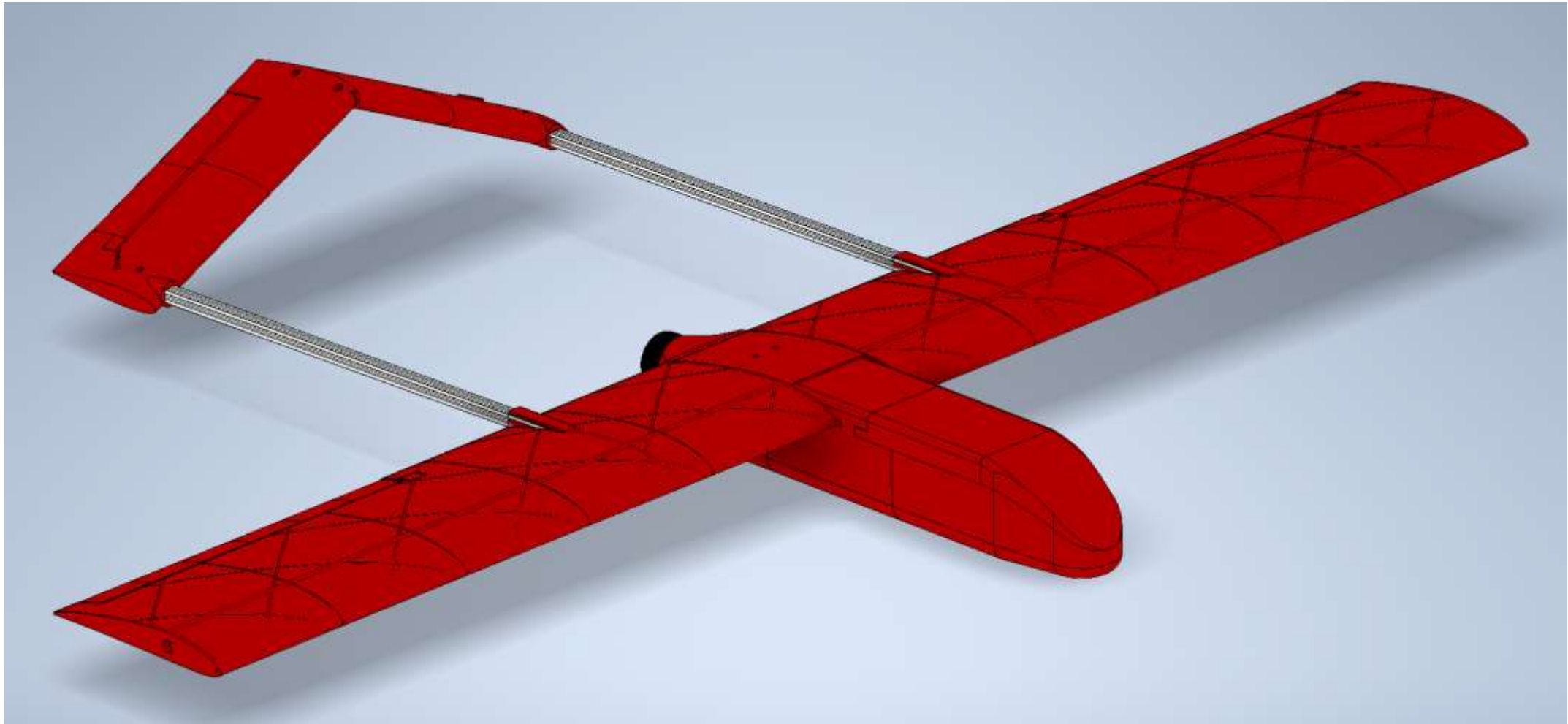
Payload Accommodation

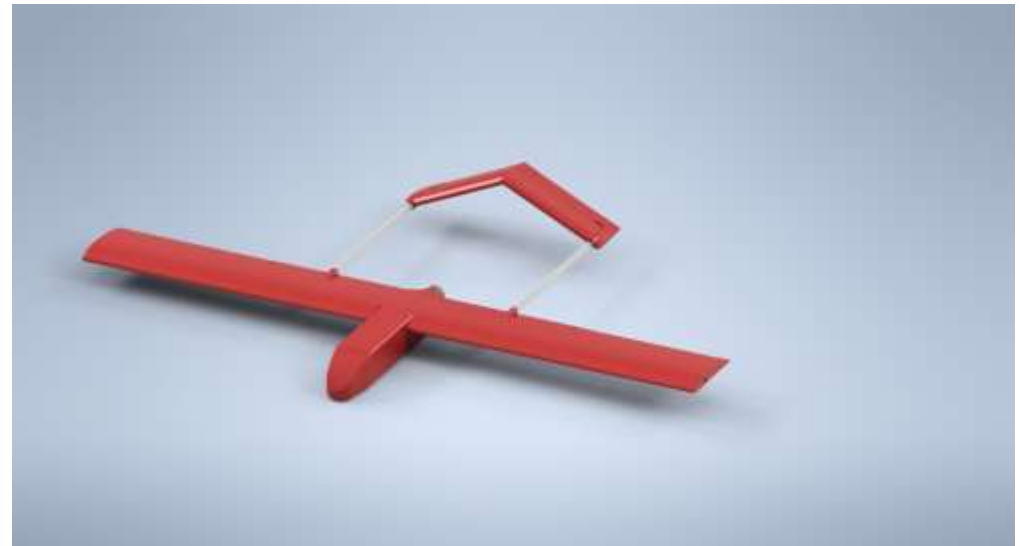
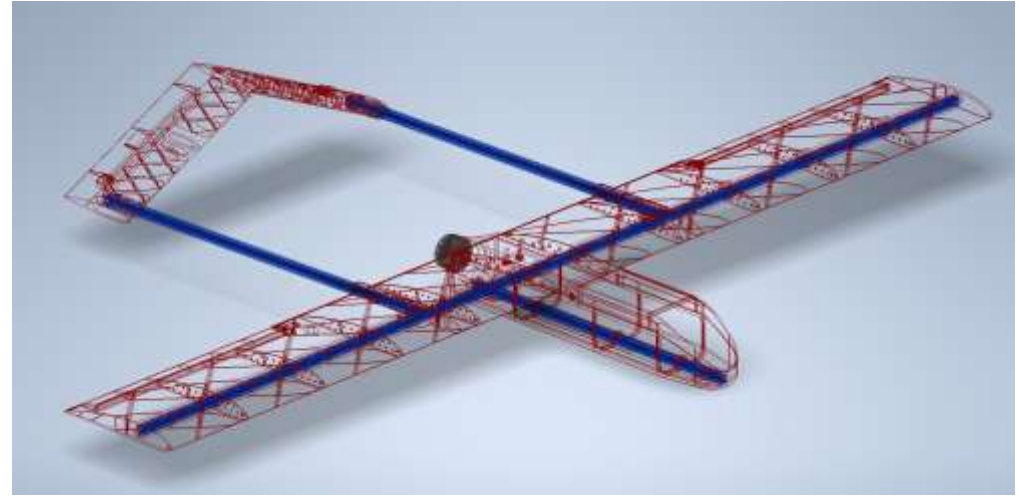
- A simple **pylon structure** as added support the mounting any of the 2" PVC chassis sensor packages the lab has developed
 - Ex. Deployable water quality sensor
- This modularity allows for **collaborative research projects**
 - external partners can attach their instrumentation without substantially altering the aircraft.
- Future upgrades may include quick-release interfaces.



Next Steps







Thank You for Your Time

Lab GitHub: github.com/arts-laboratory



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